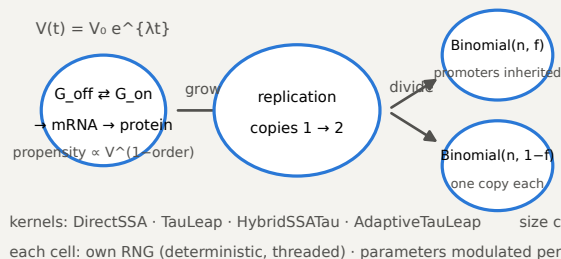


a

## Cell: reactions in a growing, dividing volume



c

## Observation and output

scRNA-seq: capture  $\sim$  Beta(mean, cv) · Poisson depth · dropout · batches; library size  $\propto$  cell volume

smFISH: fixed detection efficiency

time-lapse: reporter sampled along lineages

outputs: CSV · Newick · AnnData (.h5ad) with true volume, age, clone and copy number per cell



b

## Population · drug · lineage

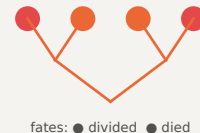
control: ConstantN (Moran replacement) · FreeGrowth · LogisticGrowth(K)

dose  $d(t)$ : constant · pulsed (holidays) · piecewise · bolus PK · function

death hazard  $h = h_{\text{max}} d^m / (EC50^m + d^m) \cdot K^q / (K^q + c_P^q)$

growth inhibition · RateModulation(param,  $f(d)$ ) · GenePerturbation

LineageTable: parent · clone · generation · birth/end time · fate · state at birth and at division  $\rightarrow$  Newick



d

## Downstream

lineage statistics: mother-daughter, sister and cousin correlations,

lineage autocorrelation, memory timescale, population vs lineage noise

fluctuation tests (Luria-Delbrück) · clonal (MemorySeq-style) scores

inference: ABC-SMC over any simulation · exact telegraph likelihood

generators: random GRNs (ER / scale-free) with known ground truth

e

```
using Biomodelling, Random
model = telegraph_model(k_on = 0.005, k_off = 0.005, k_tx = 30.0, k_dm = 1.0, k_tl = 4.0, k_dp = 0.2) # resistance gene R → P
x0 = initial_state(model; G_off = 1)
pert = Perturbation(PiecewiseDose([0.0, 100.0], [0.0, 1.0])); # drug from t = 100
effects = [DeathHazard(h_max = 0.5, EC50 = 0.3, protect = :protein, K = 60.0, q = 4.0)]
st = PopulationSettings(dt = 0.1, growth = ExponentialGrowth(log(2)/20), size_control = Sizer(2.0; cv = 0.05),
    replication = Replication(0.5), control = FreeGrowth(max_cells = 20_000))
res = simulate_population(model, x0, 1000, (0.0, 200.0); settings = st, perturbation = pert, rng = Xoshiro(1))
heritability(res, :protein; relation = :mother_daughter) # memory across divisions
Y = sequence(sample_cells(res; n = 500).counts, SeqProtocol(capture = 0.15)).Y # synthetic scRNA-seq
```